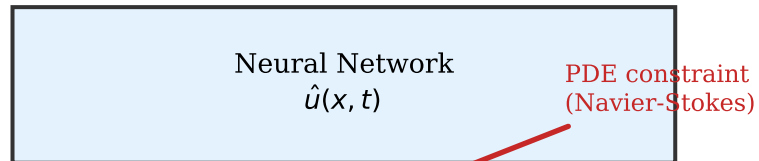


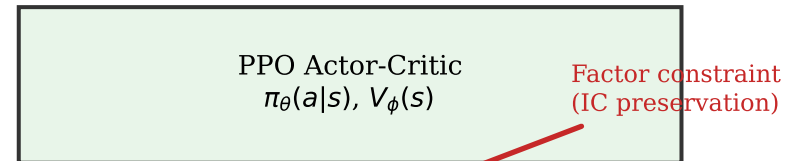
Figure 8: PINN-FI-PPO Analogy

Physics-Informed NN (PINN)



$$\text{Loss} = \mathcal{L}_{\text{data}} + \lambda_{\text{PDE}} \cdot \mathcal{L}_{\text{PDE}}$$
$$\mathcal{L}_{\text{PDE}} = \|\mathcal{N}[\hat{u}] - f\|^2$$

FI-PPO (This Work)



$$\text{Loss} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{IC}} \cdot \mathcal{L}_{\text{IC}} + \lambda_{\text{ortho}} \cdot \mathcal{L}_{\text{ortho}}$$
$$\mathcal{L}_{\text{IC}} = \text{ReLU}(-\text{IC} - \tau)^2$$